Measuring the Speed of Sound

Exam code: 0625 & 0972

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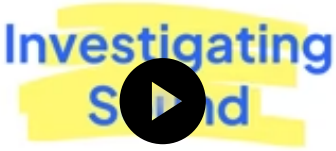
Written by: Katie M
Reviewed by: Caroline Carroll
Updated on 5 May 2025



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Speed of sound in air

- The speed of sound in air varies from **330 – 350 m/s**
  - The **higher** the air **temperature**, the **greater** the **speed** of sound

Speed of sound in materials

Extended tier only

- Sound travels at different speeds in different media:
  - Sound travels **fastest in solids**
  - Sound travels **slowest in gases**
- Sound travels at:
  - Around 1500 m/s in liquids
  - Around 5000 m/s in solids

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Measuring the speed of sound

Equipment

Trundle Wheel	To measure the distance travelled by the sound waves
Wooden Blocks	To create a sound when banged together
Stopwatch	To time how long it takes the sound waves to travel
Oscilloscope	To display the sound wave electronically
Microphones x2	To detect sound waves and turn them into an electrical signal
Tape Measure	To measure the distance between microphones

- **Resolution** of measuring equipment:
  - Trundle wheel = 0.01 m
  - Tape measure = 0.1 cm
  - Stopwatch = 0.01 s

Experiment 1: measuring the speed of sound between two points

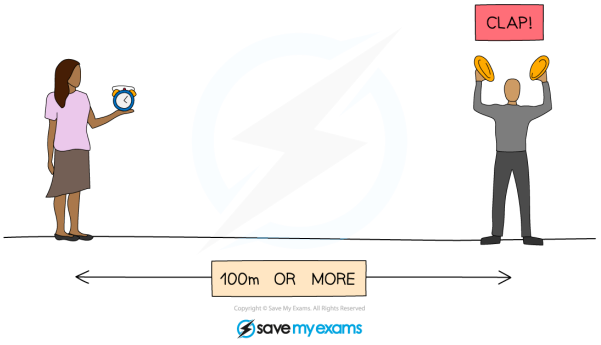
- This experiment aims to measure the speed of sound in air between two points

Variables

- **Independent variable** = Distance
- **Dependent variable** = Time
- Control variables:
  - Same location to carry out the experiment

Method

Measuring the speed of sound in air



Measuring the speed of sound directly between two points

1. Use the trundle wheel to measure a distance of 100 m between two people

3. The second person should have a stopwatch which they start when they see the first person banging the blocks together and stop when they hear the sound
4. This should be repeated several times and an average taken for the time travelled by the sound waves
5. Repeat this experiment for various distances, e.g. 120 m, 140 m, 160 m, 180 m

Results

An example results table for the speed of sound in air

Distance / m	Time 1 / s	Time 2 / s	Time 3 / s	Average time / s
100				
120				
140				
160				
180				

Analysis of results

- The speed of sound can be calculated using the equation:

$$\text{average speed} = \frac{\text{distance moved}}{\text{time taken}}$$

- The speed of sound in the air should work out to be about 340 m/s

Experiment 2: measuring the speed of sound with oscilloscopes

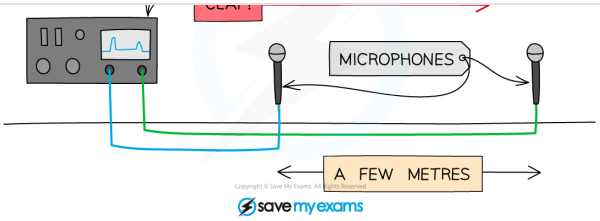
- This experiment aims to measure the speed of sound in air between two points using an oscilloscope

Variables

- **Independent variable** = Distance
- **Dependent variable** = Time
- Control variables:
  - Same location to carry out the experiment
  - Same set of microphones for each trial

Method

Measuring the speed of sound with an oscilloscope



Measuring the speed of sound using an oscilloscope

1. Connect two microphones to an oscilloscope
2. Place them about 2 m apart using a tape measure to measure the distance between them
3. Set up the oscilloscope so that it triggers when the first microphone detects a sound, and adjust the time base so that the sound arriving at both microphones can be seen on the screen
4. Make a large clap using the two wooden blocks next to the first microphone
5. Use the oscilloscope to determine the time at which the clap reaches each microphone and the time difference between them
6. Repeat this experiment for several distances, e.g. 2 m, 2.5 m, 3 m, 3.5 m

Results

An example results table for obtaining the speed of sound using an oscilloscope

Distance / m	Time 1 / s	Time 2 / s	Time 3 / s	Average time / s
2.0				
2.5				
3.0				
3.5				
4.0				

Analysis of results

- The speed of sound can be calculated using the equation:

$$\text{average speed} = \frac{\text{distance moved}}{\text{time taken}}$$

- The speed of sound in the air should work out to be about 340 m/s

Evaluating the experiments

Systematic Errors:

ensure this is taken into account when calculating speed

Random errors:

- The main cause of error in experiment 1 is the measurement of time
  - Ensure to take repeat readings when timing intervals and calculate an average to keep this error to a minimum
  - Maximise the distance between the two people where possible. This will reduce the error in measurements of time because the time taken by the sound waves to travel will be greater

Examiner Tips and Tricks

When answering questions about methods to measure waves, the question could ask you to comment on the accuracy of the measurements.

In the case of measuring the speed of sound:

- Experiment 2 is the **most** accurate because the timing is done automatically
- Experiment 1 is the **least** accurate because the time interval is very short

Whilst this may not be too important when giving a method, you should be able to explain why each method is accurate or inaccurate and suggest ways of making them better (use bigger distances)

- For example, if a manual stopwatch is being used there could be variation in the time measured which can be up to 0.2 seconds due to a person's reaction time
- The time interval could be as little as 0.3 seconds for sound travelling in the air
- This means that the variation due to the stopwatch readings has a big influence on the results and they may not be reliable

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